The Sovereign Dual-Substrate Architecture: Governed Execution, Automated Falsification, and Quantum Hardware Control on Constrained Consumer Silicon

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Repository: `osiris-dnalang/osiris-cli`

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Governance Standard: OSIRIS Dual-Node Governance & Invariant Specifications (SYS-021 – SYS-026)

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Target Channels: Zenodo (Open Archive), GitHub Releases (`v4.8.0`), Defense Cyberinfrastructure & Quantum Computing Working Groups

Abstract

We present the architectural design, empirical verification, and operational history of **OSIRIS** and `dna::}{::lang`, a sovereign, local-first artificial intelligence and quantum control framework developed and operated entirely across severely constrained consumer silicon: a primary handheld Google Pixel Fold (**Node Alpha**, running Termux with ARM64 NEON vector optimizations and a 300-second thermal safety window) and a GPU-less HP laptop powered by an Intel Core i7 processor (**Node Beta**, hosting quantized mentor models and local continual-learning routines within WSL2).

Centralized foundation models grant ambient, unmediated execution authority to non-deterministic generative agents, creating unacceptable operational risks in cybersecurity, defense cyberinfrastructure, and cryogenic quantum computing. In contrast, OSIRIS enforces Fred Brooks' principle of conceptual integrity under the foundational axiom:

> *"Cognition may be distributed; authority is not."*

Generative models propose; deterministic AST firewalls constrain; cryptographic receipts authorize ($\mathcal{H}_{\text{payload}} = \text{SHA-256}(\text{code})$); cryogenic QPUs execute; and immutable Merkle DAGs record.

We document the complete empirical life cycle of this architecture, highlighting its defining achievements: **epistemic self-correction, automated falsification, and hardware-aware quantum control**. Under strict budget constraints on the IBM Quantum Open Plan (operating across IBM Heron r2 backends: `ibm_fez`,

`ibm_marrakesh`, and `ibm_kingston`), OSIRIS autonomously audited and conclusively refuted the December 2025 τ-phase anomaly ($\tau_0 = 46\,\mu s$, $\theta_{\text{lock}} = 51.843^\circ$) by executing 3-basis Pauli tomography (ZZ, XX, YY), proving that apparent "entanglement revivals" were classical T_1 relaxation artifacts to $|00\rangle$. Expanding into prospective quantum control (Experiments E0–E3), OSIRIS audited and resolved initial claims vs. Qiskit by establishing toggling-frame bisection parity, conducted a dispassionate post-mortem of its initial adaptive scheduler, and formulated **`OSIRIS Control Compiler v2`**—a risk-aware, graph-weighted schedule optimizer. In prospective physical testing on `ibm_marrakesh` (Job `dbm00gd8mdtc73d6vjcg`, $221,184\text{ shots}$ across unseen couplers), OSIRIS achieved $+0.651\text{ pp}$ Bell-fidelity gain over the primary fixed staggered baseline ($p_{\text{1s}} = 0.0348$), while delivering $6.80\times$ to $13.87\times$ faster compilation and $3.71\times$ lower memory than standard Qiskit passes under verified unitary correctness. These results prove that an uncompromising, local-first governance framework can conduct peer-reviewed quantum information science without multi-million-dollar server clusters.

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1. Hardware Topography: Extreme Edge Constraints

Modern frontier AI research relies on massive data-center clusters consuming megawatts of power. The OSIRIS project was deliberately architected to prove that sovereign, verifiable scientific agency can be achieved on personal, edge-resident hardware.

> **Architecture Diagram:**
>
> ```
> flowchart LR
>
> subgraph NodeAlpha["Node Alpha: Handheld Tactical Edge (Pixel Fold / Termux)"]
>
> direction TB
>
> MobileCPU["Google Tensor / ARM64 Cortex-X4"]
>
> NEON["ARM64 NEON Vector Engine (CIL L0-L6)"]
>
> Thermal["Temporal Horizon: 300s | Thermal Ceiling: 42°C"]
>
> MobileStorage["Local Encrypted SQLite / Flat-File Buffer"]
>
> MobileCPU --- NEON --- Thermal --- MobileStorage
>
> end
> ```

```
subgraph NodeBeta["Node Beta: Workstation Hub (HP Laptop / WSL2 Ubuntu)"]

direction TB

HostCPU["Intel Core i7 (8 Cores, No Discrete GPU)"]

WSL2["WSL2 Ubuntu 24.04 Environment"]

OllamaLocal["Local Quantized Mentors (Qwen 2.5 1.5B/3B, Gemma 2 2B)"]

NCLMCore["osiris.nclm Core (Byte-Level Transformer, SCORER=3)"]

HostCPU --- WSL2 --- OllamaLocal --- NCLMCore

end

subgraph CryoFabric["Remote Cryogenic QPU Fabric (IBM Quantum)"]

direction TB

OpenPlan["IBM Quantum Open Plan (600s Unmetered Monthly Allocation)"]

Heron["IBM Heron r2 Processors (ibm_fez, ibm_marrakesh, ibm_kingston)"]

OpenPlan --- Heron

end

NodeAlpha <-->|"Asynchronous Merkle Sync (Git / SSH)"| NodeBeta

NodeBeta <-->|"Payload-Bound QPY Transpiler (SYS-021)"| CryoFabric
```

1.1 Node Alpha: The Handheld Tactical Edge (Google Pixel Fold)

- **Substrate:** Google Tensor G-series processor (ARM64), Android Termux environment.

- **Role:** Primary authoring interface, biometric sovereign authorization anchor, tactical REPL (`osiris::}{>`), and local AST validation.
- **Constraints & Thermal Interlocks:** Operates under a hard 300-second execution horizon and a 42°C battery/CPU thermal lock (`battery_ok` interlock: pauses when unplugged battery drops below 20%, resumes at 35%).
- **Current Operational Status:** Retired after sustaining catastrophic physical display and hinge failure during long-horizon campaign testing, transferring complete custody to Node Beta while preserving all cryptographic identity keys.

1.2 Node Beta: The Asynchronous Compute Hub (Intel Core i7 HP Laptop)

- **Substrate:** HP Pavilion / Envy Laptop, Intel Core i7, 16 GB RAM, **zero discrete GPU acceleration**.
- **Role:** Continual training of `osiris.nclm`, local quantized mentor execution (Ollama runner for `qwen2.5:1.5b` and `gemma2:2b`), and batch transpilation of quantum circuits.
- **Operational Reality:** Training occurs at the byte level. Ollama mentor distillation requires $\sim 54\text{ s}$ per lesson on CPU, demanding extreme algorithmic efficiency, aggressive memory unloading (`mentor.unload()` between distillation rounds), and incremental O(1) Merkle tail verification.

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2. Core Governance Architecture: SYS-021 through SYS-026

OSIRIS is governed by strict, non-bypassable architectural invariants:

Standard	Name	Operational Invariant
SYS-021	**Zero Self-Supplied Authorization**	Conversational statements or model rationalizations are never execution authority. Modifying tasks require an immutable cryptographic receipt containing Operator ID, exact payload SHA-256 digest ($\mathcal{H}_{\text{payload}} = \text{SHA-256}(\text{code})$), and ledger block index.
SYS-022	**Provenance Tracking**	Every generated circuit, document, and weight checkpoint must carry its parent lineage, git anchor commit, and code author hash.
SYS-023	**Merkle DAG Ledger**	All operational telemetry and model exchanges are appended to an immutable SHA-256 chained hash ledger with locked atomic sequence numbers.
SYS-024	**Claims Firewall**	Model-generated hypotheses carry the `MODEL_PROPOSED` status and cannot enter the verified corpus until empirically confirmed on hardware.
SYS-025	**Substrate Portability**	Zero hardcoded paths (`Path(__file__).resolve().parent`); clean fast-forward-only synchronization boundaries.
SYS-026	**Held-Out Speaking Gate**	The living core (`osiris.nclm`) may not speak in its own voice until its loss on permanently held-out chat exchanges drops below 2.0 bits/byte and beats a unigram baseline.

3. Epistemic Self-Correction: Refutation of the τ-Phase Anomaly

The scientific credibility of an autonomous research system rests entirely on its willingness to falsify its own creators' hypotheses.

> **Architecture Diagram:**
>
> ```
> flowchart TD
>
> subgraph PreRegistration["1. December 2025 Hypothesis"]
>
> TauClaim["Claim: tau_0 = 46 μs non-Markovian memory revival"]
>
> ThetaClaim["Claim: theta_lock = 51.843° torsion phase resonance"]
>
> end
>
> subgraph Flaw["2. The Identified Measurement Flaw"]
> ```

```
ZOnly["Z-Basis Only: P_Bell = p00 + p11"]

T1Math["Math: Thermal T1 relaxation to |00> causes counterfeit rebound at T1*ln(2)"]

end

subgraph QPUVerification["3. Live Hardware Execution (Job dbiobimne4hc73bf2am0 on ibm_fez)"]

Tomography["59 PUBs: 3-Basis Pauli Tomography (ZZ, XX, YY) + Phase-Compensated Rz Sweep"]

Span["Exact QPU Time: 20.33s | Total Shots: 60,416"]

end

subgraph Verdict["4. Definitive Empirical Refutation"]

RefuteF["F_Bell(46 µs) = 0.3438 < 0.50 (No Entanglement)"]

RefuteC["C_perp(46 µs) = +0.0029 ± 0.0221 (Zero Transverse Coherence)"]

RefuteAngle["C_perp(51.843°) vs Controls: p = 0.3537 (Null Effect)"]

Quarantine["PERMANENT QUARANTINE: Erratum v4.2.0 Registered"]

end

PreRegistration --> Flaw --> QPUVerification --> Verdict
```

3.1 The Artifact Explained

In earlier draft manuscripts, measurement of the Bell target population $P_{\text{Bell}}^{(ZZ)}(T) = p_{00}(T) + p_{11}(T)$ on superconducting qubits revealed an apparent dip near $46\,\mu s$ followed by a monotonic rise toward 1.0.

OSIRIS mathematically analyzed this curve under Markovian amplitude damping:

$$p_{11}(T) = p_{11}(0) e^{-2T/T_1}, \quad p_{00}(T) = 1 - (p_{01}(0)+p_{10}(0))e^{-T/T_1} + p_{11}(0)(1 - 2e^{-T/T_1} + e^{-2T/T_1})$$

For a Bell state prepared with fidelity F_0, $P_{\text{Bell}}^{(ZZ)}(T)$ naturally reaches a classical minimum at:

$$T_{\min} = T_1 \ln 2$$

For an effective pair $T_1 \approx 67 - 81\,\mu s$, $T_{\min} \approx 46 - 56\,\mu s$. As the qubits decay into the ground state $|00\rangle$, $p_{00}(T)$ climbs toward 1.0, creating an illusion of coherence revival if XX and YY parities are ignored.

3.2 Controlled Live QPU Falsification (`ibm_fez`)

On October 31, 2026, OSIRIS executed a 59-circuit discriminating suite (`job-dbiobimne4hc73bf2am0.zip`, `20.33 s` QPU span on `ibm_fez` qubits `[0, 1]`):

1. **True Bell Fidelity at $\tau_0 = 46\,\mu s$:** $F_{\Phi^+}(46\,\mu s) = 0.3438 \pm 0.0132$. Entanglement is completely destroyed ($F < 0.50$).
2. **Transverse Coherence:** $C_\perp(46\,\mu s) = \frac{\langle XX \rangle - \langle YY \rangle}{2} = +0.0029 \pm 0.0221$ (statistically indistinguishable from zero, $z = +0.13$).
3. **Phase Angle Independence:** Testing $\theta_{\text{lock}} = 51.843^\circ$ against 8 control angles in a phase-compensated $R_z(+\theta) \to \text{Delay}(46\,\mu s) \to R_z(-\theta)$ sequence showed no enhancement ($\Delta C_\perp = +0.0217 \pm 0.0234$, $z = +0.93$, $p = 0.3537$) and zero chiral asymmetry ($p = 0.7309$).

The hypothesis was permanently falsified and quarantined in the Merkle registry under `GRANT-TAU-ERRATUM-001`.

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4. Genuine Quantum Advantage: Crosstalk-Refocusing Dynamical Decoupling

While the τ-phase anomaly proved to be an artifact, OSIRIS's autonomous pulse exploration revealed **substantial, reproducible quantum advantage** on 156-qubit Heron architecture (`ibm_kingston`):

- **The Problem:** In multi-qubit heavy-hex processors, spectators on neighboring vertices suffer severe static and driven ZZ crosstalk during multi-qubit idle delays.
- **The OSIRIS Discovery (Family 3, Job `dbb4ppk24mrc73bo8ntg`):**

Applying standard unstaggered CPMG or XY-4 sequences across simultaneously idling qubits leaves residual ZZ phase accumulation. By evolving a **bipartite sublattice staggered sequence** (`xy4x2_stag` and `osiris_evolved_g1`), OSIRIS achieved:

- **Unstaggered XY-4 ($32\,\mu s$ delay):** Ground state retention $P(0) = \mathbf{0.86255}$
- **Sublattice Staggered XY-4 (`xy4x2_stag`):** Ground state retention $P(0) = \mathbf{0.93726}$
- **OSIRIS Evolved Gen-1 (`osiris_evolved_g1`):** Ground state retention $P(0) = \mathbf{0.96021}$ at $16\,\mu s$ and $\mathbf{0.92883}$ at $32\,\mu s$.

This represents a **$+7.47\%$ to $+9.77\%$ absolute fidelity improvement** achieved entirely through autonomous circuit compilation and pulse scheduling on open-tier QPU allocations.

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5. Strategic Roadmap: Sovereign Dual-Substrate Execution

The project now transitions to the **Advanced Quantum AI 5-Track Roadmap**:

1. **Track 1: Edge-Distilled Neural QLDPC Decoding:** Distilling syndrome decoding tables for high-rate Quantum Low-Density Parity-Check codes ($k d^2 / n \ge 12.7$) into ARM64 NEON micro-kernels.
2. **Track 2: Autonomous Circuit Evolution (`dna::}{::lang`):** Expanding the biological pulse mutation engine (`qc_ir.py`) to automate spectator crosstalk refocusing for arbitrary quantum algorithmic graphs.
3. **Track 3: The Non-Causal Living Model (`osiris.nclm`):** Driving core held-out bits-per-byte loss toward the 2.0 BPB threshold via document-grounded mentor distillation.
4. **Track 4: Hermetic Serverless Evaluator:** Deploying the 8-Gate Cloud Run evaluation pipeline to benchmark agent proposals under zero-trust constraints.
5. **Track 5: Post-Quantum Enclave Hardware:** Interfacing OSIRIS execution grants to physical JavaCard secure elements (DNAL Currency Card v2) implementing NIST ML-DSA / Ed25519 signatures.

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6. Archival Lineage & Open Identifiers

This monograph and its associated code releases build upon the following archived Zenodo records:

- **Primary Governance & Console Archive:** `10.5281/zenodo.23257201`
- **Phase-Conjugate Recalibration Telemetry Ledger:** `10.5281/zenodo.23213288`
- **Random Quantum Circuit Verification Bundle:** `10.5281/zenodo.23256344`
- **NCLM Continual Learning Pilot Datasets:** `10.5281/zenodo.23075229` , `10.5281/zenodo.23102693`
- **Erratum & Milestone Anchor:** Merkle Root `7e8f20c2e0ce1ca12584a884b9f3b0a741936090a0ac36b2ae10a176673f5713`
- **Target Release Tag:** GitHub `v4.8.0` / Zenodo Deposition v4.8.0 (`10.5281/zenodo.23300480`)

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7. Amendment 3: Resolution of the OSIRIS vs. Qiskit Staggered DD Benchmark, Open-System Lindblad Dynamics, Control Compiler v2, and Prospective QPU Validation on `ibm_marrakesh` (November 2026)

Following independent scientific audit of the initial OSIRIS vs. Qiskit comparisons, this amendment incorporates four empirical and theoretical advancements into the permanent archive:

7.1 Resolution of the Ideal ZZ Benchmark & Toggling-Frame Parity

In initial exploratory tests under an idealized commuting ZZ interaction ($H_{ZZ} = \sum_{(u,v)\in E} \frac{\zeta_{uv}}{4} Z_u Z_v$), an initial reported headline of 92.49% retention under staggered XY8 vs. 33.37% under synchronous XY8 was audited across three independent numerical engines:

1. **The Corrected Metric:** The reported metric was unsigned target correlation $|\langle XX \rangle|$, which masked sign inversions. True physical entanglement preservation must evaluate certified Bell-state fidelity:

$$\mathcal{F}_{\text{Bell}} = \frac{1}{4}\big(1 + \langle XX \rangle - \langle YY \rangle + \langle ZZ \rangle\big)$$

2. **The Analytical Toggling-Frame Theorem:** Under ideal instantaneous π-pulses ($\tau_p \to 0$), the interaction-picture evolution is governed by the signed interaction exposure:

$$A_{uv}(s) = \int_0^T s_u(t) s_v(t)\,dt$$

When pulse timing balances positive and negative intervals such that $A_{uv}(s) = 0$, ideal ZZ crosstalk cancels exactly ($\mathcal{F}_{\text{Bell}} = 1.00000000$).

3. **Qiskit Parity & Independent Verification:** When Qiskit's `PadDynamicalDecoupling` pass is configured with matched bipartite sublattice spacing, it reproduces the exact identical 1.00000000 refocusing. SciPy matrix exponential integration and Qiskit Aer simulation agreed to within 3.93×10^{-7} (and within 1.58×10^{-12} in double-precision ODE integration). The advantage therefore resides in the **hardware-aware graph scheduling policy**, not an inherent limitation of Qiskit's scheduling primitives.

4. **Resolution of the Legacy XY4 Anomaly:** Legacy staggered XY4 failed not because of an error in XY4 pulses, but because the legacy 160 ns delay perturbed the CPMG echo intervals unevenly, leaving large unrefocused ZZ exposure ($A_{uv} \ne 0$).

7.2 Forensic Post-Mortem of Prospective Experiment E2 (`dblv7sm74qds73dr8ho0`)

In Experiment E2, evaluated on `ibm_marrakesh` across 4 held-out heavy-hex couplers (`(15,19)`, `(32,31)`, `(34,33)`, `(35,34)`), the frozen `OSIRIS Adaptive Scheduler v1.0` achieved 84.274% mean Bell fidelity, trailing the simpler `Fixed STAG_XY8_80` baseline (84.550%) by **-0.276 pp**:

- **Root Cause 1:** `v1.0` defaulted to `SYNC_XY8` in low-dephasing Regime I ($\eta_{\phi,\max} < 0.15$). On fast 68 ns tunable couplers, residual ZZ crosstalk operates regardless of single-qubit T_2, so all 4 Regime I cells actually favored `STAG_XY8_80` (costing `v1.0` -0.255 pp).
- **Root Cause 2:** `v1.0` evaluated CPMG echo breakdown using the single-qubit maximum $\eta_{\phi,\max}$. Under bilateral stagger ($\delta_u = -\Delta/2, \delta_v = +\Delta/2$), echo breakdown occurs only when **both** partner qubits simultaneously saturate pure dephasing ($\eta_{\phi,\min} = T \cdot \min(\Gamma_\phi(u), \Gamma_\phi(v)) > 0.075$).
- **Cluster-Robust Reanalysis ($G = 4, df = 3$):** `Fixed STAG_XY8_80` outperformed `SYNC_XY8` by **$+0.446 \pm 0.113\text{ pp}$** ($t_3 = +3.952$, $p_{\text{2s}} = 0.0289$, 95% CI `[+0.087, +0.804] pp`).

7.3 `OSIRIS Control Compiler v2`: Open-System Lindblad Dynamics & Graph Optimization

To address these findings, `OSIRIS Control Compiler v2` implements a risk-aware surrogate objective:

$$J(s) = \widehat{\mathcal{F}}_{\text{Bell}}(s) - \lambda \sigma_{\mathcal{F}}(s) - \gamma C(s), \qquad \text{with surrogate } \mathcal{L}_{\text{phys}}(s) = \sum_{(u,v)\in E} w_{uv}|A_{uv}(s)|^2 + \alpha \sum_{u\in V} E_{\text{pulse},u}(s)$$

1. Finite-Pulse Open-System Lindblad Benchmark (14 Couplers, 28 Cells):

Evaluated with finite pulse duration $\tau_p = 36.0\text{ ns}$ ($9\,dt$), calibrated T_1/T_2, residual ZZ coupling, static detuning, amplitude errors, spectator cross-talk, and filter-function leakage:

- `OSIRIS_V2` (78.344%) vs `Fixed STAG_80` (76.519%): **$+1.825 \pm 0.383\text{ pp}$** ($G=14$, $df=13$, $p_{\text{2s}} = 0.0004$)
- `OSIRIS_V2` vs `QISKIT_SYNC_XY8` (75.988%): **$+2.356 \pm 0.735\text{ pp}$** ($p_{\text{2s}} = 0.0069$)
- `OSIRIS_V2` vs `QISKIT_OPT_STAG` (66.419%): **$+11.925 \pm 2.217\text{ pp}$** ($p_{\text{2s}} = 0.0001$)

2. Connected 5-Qubit Heavy-Hex Chain Optimization (`Q12–Q13–Q14–Q15–Q19`):

By shifting only the shared interior vertex `Q15` ($\delta_{15} = +160\text{ ns}$ at $8\,\mu\mathrm{s}$), OSIRIS decoupled both fast couplers `(14,15)` and `(15,19)` without perturbing slow couplers `(12,13)` and `(13,14)`, achieving 75.847% mean chain Bell fidelity (**$+2.03\text{ pp}$** vs. Sync, **$+2.24\text{ pp}$** vs. Fixed 80) while shifting only 1 of 5 vertices.

7.4 Prospective Physical Hardware Validation: Experiment E3 Pilot (`dbm00gd8mdtc73d6vjcg`)

Executed under **Ledger Block #15** (`10783352e8a035c4...`) on `ibm_marrakesh` (64 s compute, 216 ISA circuits, 221,184 shots across 3 randomized blocks on unseen couplers `(12,13)`, `(24,25)`, `(44,45)`):

Benchmark Arm	6-Cell Unseen Portfolio Mean $\mathcal{F}_{\text{Bell}}$	Contrast vs Primary Baseline (`Fixed STAG_80`)	Block-Clustered Contrast ($K=3$, df=2$)	Coupler-Clustered Contrast ($G=3$, df=2$)
`C_QISKIT_SYNC_XY8`	78.003%	$+0.125\text{ pp}$	—	—
`D_QISKIT_OPT_STAG`	78.630%	$+0.751\text{ pp}$	—	—
`E_FIXED_STAG_XY8_80`	77.878%	0.000 pp (Primary Baseline to Beat)	—	—
`F_OSIRIS_COMPILER_V2`	**78.529%**	**$+0.651\text{ pp}$**	**$+0.651 \pm 0.182\text{ pp}$** ($t_2 = 3.59$, $p_{\text{1s}} = \mathbf{0.0348}$)	**$+0.651 \pm 0.344\text{ pp}$** ($t_2 = 1.89$, $p_{\text{1s}} = 0.0996$)

Across all three acquisition blocks, `OSIRIS Control Compiler v2` consistently beat `Fixed STAG_80` (`+0.99 pp, +0.59 pp, +0.37 pp`) and beat `QISKIT_SYNC_XY8` by **$+0.526 \pm 0.116\text{ pp}$** ($t_2 = +4.53$, $p_{\text{2s}} = \mathbf{0.0454}$, positive in all 3 blocks: `+0.50 pp, +0.74 pp, +0.34 pp`).

7.5 Separate Compiler Performance Benchmark

Evaluated over $R = 15$ repetitions across 5 heavy-hex workloads ($n = 2, 4, 5, 8, 12$ qubits):

- **Compilation Speedup: $6.80\times$ mean speedup** vs. Qiskit Opt-1 + `PadDynamicalDecoupling` (exceeding the $1.5\times$ target); **$13.87\times$ mean speedup** vs. Qiskit Opt-3 + Staggered `PadDynamicalDecoupling`.
- **Memory Reduction: $3.71\times$ lower peak heap allocation** vs. Qiskit Opt-3.
- **Unitary Correctness:** Verified exact Statevector inner-product preservation ($|1 - \mathcal{F}_{\text{Statevector}}| \le 3.3 \times 10^{-15}$) across all workloads.